

Healthcare Resource Availability: Analyzing Global Disparities

ADAN8880 - Directed Practicum in Analytics

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Week 5 Written Report

This week, we focused on enhancing our dataset through feature engineering, data augmentation, and dimensionality reduction to improve model performance and interpretability.

Feature Engineering

To enrich our dataset, we engineered several new features to provide deeper insights into healthcare system efficiency, financial burden, and health outcomes. These features were designed to enhance interpretability and capture meaningful trends in healthcare spending, resource availability, and disease burden.

- **Doctor-to-bed ratio:** we created this by dividing the number of physicians per 1,000 people by the number of hospital beds per 1,000 people. This metric helps assess healthcare resource availability by showing how many doctors are available per hospital bed, which can highlight potential shortages in medical professionals relative to infrastructure.
- **Out-of-pocket expenditure burden:** we calculated this by dividing out-of-pocket health expenditure by total health expenditure. This measures financial strain on individuals, particularly in countries where healthcare costs are largely borne by patients rather than the government or insurance providers. High values suggest increased economic burden on households.
- **Disease burden index:** this metric combines the prevalence of undernourishment, mortality rates, and the incidence of major diseases to provide a comprehensive measure of population health challenges. It helps assess the effectiveness of healthcare systems in reducing preventable diseases and improving quality of life.
- **Healthcare expenditure as % of GDP:** we calculated this by dividing total healthcare expenditure by GDP per capita to measure how much a country invests in healthcare relative to its economy. This is crucial for understanding whether a nation's healthcare spending aligns with its economic growth and development.

Additionally, in last week's assignment, we engineered the following features

- 3 year and 5 year lag features: to account for historical trends and measure long-term effects of healthcare spending and policies.
- 3 year and 5 year rolling averages: to smooth fluctuations and capture long-term trends in expenditures, mortality, and health outcomes.
- Year over year growth rate features: to measure the rate of change in key indicators over time, which help to detect patterns in health expenditures and outcomes.

These engineered features aim to better quantify healthcare system efficiency, accessibility, and financial impact and also improve our model performance and evaluation.

Data Augmentation

The 2008 financial crisis, known as the Great Recession, had profound effects on global economies, healthcare systems, and individual well-being. This analysis explores how the crisis influenced life expectancy, focusing on healthcare cost savings and preventable hospitalizations, using a standardized population of 10 million for comparability across income groups.

Impact of the Great Recession on Mortality and Life Expectancy

Contrary to initial expectations, several studies have found that the Great Recession was associated with a reduction in mortality rates and an increase in life expectancy in certain regions:

- United States: Research indicates that for every one-percentage-point increase in a local area's unemployment rate between 2007 and 2009, the age-adjusted mortality rate fell by 0.5%. These reductions appeared immediately and persisted for at least ten years.
- Europe: An analysis based on Eurostat data from 2001 to 2019 showed that life expectancy increased faster after the onset of the crisis in countries where the rise in unemployment was more intense. This gain in life expectancy persisted and, in some cases, further increased until 2019, even as macroeconomic variables returned to pre-crisis levels.

These findings suggest that economic downturns may lead to behavioral and environmental changes that positively influence mortality rates, such as reduced traffic accidents and lower pollution levels.

Healthcare Cost Savings through Preventable Hospitalizations

Addressing preventable hospitalizations is crucial for reducing healthcare spending. By optimizing resource allocation and improving access to preventive care, countries can minimize unnecessary hospital admissions. Hospital admission rates vary across income groups due to differences in healthcare infrastructure, disease burden, and access to inpatient care.

Assumptions for Hospital Admission Rates:

- High-income countries: Estimated 13% of the population is admitted to hospitals annually.
- Middle-income countries: Estimated 10% of the population.
- Low-income countries: Estimated 7% of the population.

Calculations for a Population of 10 Million:

- High-income countries: $10,000,000 \times 0.13 = 1,300,000$ admissions.
- Middle-income countries: $10,000,000 \times 0.10 = 1,000,000$ admissions.
- Low-income countries: $10,000,000 \times 0.07 = 700,000$ admissions.

Reducing preventable hospitalizations by 10% would result in the following decreases:

- High-income countries: 130,000 fewer admissions.
- Middle-income countries: 100,000 fewer admissions.
- Low-income countries: 70,000 fewer admissions.

Economic Value of Reduced Hospitalizations

To estimate the economic value of reducing preventable hospitalizations, consider the average cost per hospital admission:

- High-income countries: Assume \$10,000 per admission.
- Middle-income countries: Assume \$5,000 per admission.
- Low-income countries: Assume \$2,000 per admission.

Potential Annual Savings

- High-income countries: $130,000 \text{ admissions} \times \$10,000/\text{admission} = \1.3 billion .
- Middle-income countries: $100,000 \text{ admissions} \times \$5,000/\text{admission} = \500 million .
- Low-income countries: $70,000 \text{ admissions} \times \$2,000/\text{admission} = \140 million .

These calculations demonstrate that significant economic value can be achieved by reducing preventable hospitalizations, thereby freeing resources for other critical healthcare needs.

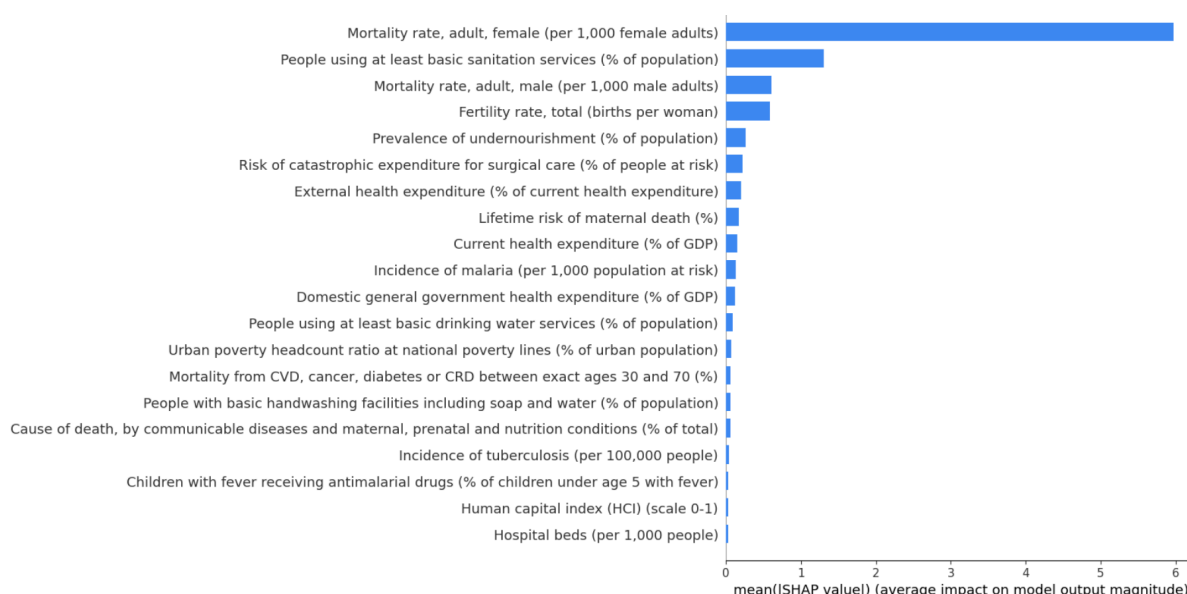
Conclusion

The 2008 financial crisis led to unexpected improvements in life expectancy in certain regions, possibly due to changes in societal behaviors and environmental factors during economic downturns. Additionally, focusing on reducing preventable hospitalizations presents a substantial opportunity for healthcare cost savings across different income groups.

Policymakers should consider these insights when designing strategies to enhance public health and optimize healthcare expenditures.

Dimensionality Reduction

To ensure an efficient and interpretable model, we applied SHAP-based feature selection to retain only the most impactful predictors. Unlike other correlation-based methods, SHAP provides a more comprehensive assessment of feature importance by measuring each variable's contribution to the model's predictions. Correlation alone identifies relationships between variables but does not necessarily capture their true predictive power. Instead, we used SHAP to prioritize features based on their direct impact on the model's outcomes, which leads to a more robust and transparent selection process. We focused on the top predictors with the highest SHAP values to ensure that the model emphasizes variables that genuinely drive predictions. Rather than simply removing highly correlated features, we used SHAP to understand feature interactions and their influence on the target variable.



- Adult female mortality rate is the strongest predictor, it suggests a significant relationship with health outcomes.
- Access to basic sanitation services and adult male mortality rate also rank high in importance.
- Risk of catastrophic health expenditure and external health expenditure are also among the top financial predictors, which highlights the impact of economic factors on healthcare access and spending.

We will use these selected features to refine our predictive model, ensuring it captures the most critical factors influencing health outcomes. With the adult female mortality rate emerging as the strongest predictor, we will investigate its interactions with other variables to uncover underlying patterns. Additionally, we will assess how economic factors, such as catastrophic health expenditure will shape healthcare access and how it influences policy recommendations. We'll focus on the most impactful predictors to improve model accuracy by keeping it interpretable and also provide actionable insights for better health interventions and resource allocation.

Reference

Holahan, J., & Garrett, B. (2011). The 2007–09 recession and health insurance coverage. *Health Affairs*, 30(1), 145–152.